PROPOSTA DI GRANT GIOVANI IN CSN5 - Anno 2026

Acronimo:	PRISM
Titolo completo:	Precision Room-temperature Instrumentation for high-energy x-ray absorption Spectroscopy Measurements
Proponente:	Simone Manti
Unità partecipanti:	INFN Sezione di Frascati

Abstract (2000 caratteri)

X-ray absorption spectroscopy (XAS) is a powerful element-selective technique for probing the electronic and atomic structure of materials. While synchrotron and free-electron laser facilities provide outstanding performance, their limited accessibility motivates advanced laboratory instrumentation. A major challenge is extending fluorescence-mode XAS beyond 20 keV, where silicon detectors lose efficiency and germanium detectors require cryogenic operation. This limits studies of elements such as Mo, Rh, Ag and Cd, whose K-edges are relevant to catalysis, electrochemistry, advanced alloys and energy materials. PRISM will develop a high-resolution laboratory hard X-ray spectrometer combining advanced crystal optics with a multichannel cadmium zinc telluride (CZT) detector for precision fluorescence spectroscopy up to 50 keV. Building on my expertise in crystal-based X-ray spectroscopy gained within the VOXES project and SIDDHARTA-2 experiment, PRISM will develop detector architectures, low-noise front-end electronics, calibration procedures and spectral reconstruction algorithms to achieve sub-keV energy resolution, high quantum efficiency (>90% up to ~40 keV) and excellent SNR performance at room temperature. The instrument will be validated through fluorescence XAS measurements on reference materials with K-edges above 20 keV, demonstrating accurate edge reconstruction and XANES spectra within laboratory-compatible acquisition times. The resulting capability will enable extended in situ and operando studies of slowly evolving catalytic, electrochemical and energy-related processes. Beyond this proof of concept, PRISM will establish a new detector platform for next-generation laboratory hard X-ray spectroscopy, strengthening INFN expertise in semiconductor radiation detectors, precision X-ray instrumentation and crystal spectroscopy. The technology will complement future photon-source infrastructures such as EuAPS and support research and industrial applications.

1 State of the art (4000 caratteri)

A central challenge in materials characterization is determining how electronic and local atomic structure evolve under realistic conditions. Changes induced by irradiation, processing, phase transformations, chemical reactions or device operation may be transient, reversible or environment-dependent, and can therefore be missed by ex-situ measurements [1]. This requires in-situ and/or operando techniques capable of probing thick, heterogeneous or diluted samples while retaining sensitivity to selected elements and their local coordination [2,3]. Hence, there is a broad need for non-destructive, element-selective and bulk-sensitive probes compatible with complex environments and long acquisition campaigns.

X-rays are particularly suited for such investigations because of their high penetration depth with respect to other optical techniques, enabling access to the bulk of samples. By tuning the photon energy of a selected element, X-ray spectroscopies provide element- and orbital-specific information, even in chemically complex and heterogeneous systems. These properties make X-ray methods well-suited to non-destructive in situ and operando studies [4].

Among X-ray techniques, X-ray absorption spectroscopy (XAS) is particularly suited to this purpose because it measures the energy-dependent absorption coefficient across an element-specific absorption edge [5,6]. The near-edge region, the XANES, is sensitive to oxidation state, electronic structure and coordination geometry, whereas the extended region up to 1 keV above the edge, the EXAFS, provides quantitative information on the local atomic environment, including interatomic distances, coordination numbers and structural disorder. XAS can be measured in transmission, by recording the attenuation of the incident beam through the sample, or in fluorescence, by detecting the characteristic X-rays emitted after absorption. While transmission is generally preferred for homogeneous samples of suitable thickness, fluorescence detection is essential for diluted absorbers, thick specimens and complex sample environments, where the transmitted intensity is too weak or difficult to access.

Synchrotron facilities are the reference platforms for XAS, as their high brilliance, broad energy tunability and dedicated beamlines enable high-quality spectra with excellent signal-to-noise ratio over a wide energy range. X-ray free-electron lasers further provide ultrashort and intense X-ray pulses, opening access to femtosecond time-resolved experiments and transient states. However, access to these large-scale facilities is limited and competitive, and the allocated beamtime is often unsuitable for routine characterization, long experimental campaigns, or studies involving hazardous and radioactive samples [7].

These limitations have driven renewed interest in laboratory XAS systems based on conventional X-ray tubes [8]. Although laboratory sources provide substantially lower brilliance than synchrotrons, they offer continuous access, flexible sample handling and long measurement times, making them attractive for routine analysis, method development and slowly evolving in situ or operando processes. Recent advances in source brightness, crystal optics and detection technology have enabled an increasing number of XAS and XES measurements to be transferred back to the laboratory.

Within INFN, VOXES [9] has established expertise in high-resolution von Hamos spectroscopy for extended and dilute X-ray sources. The instrument currently operates mainly in the 5–20 keV range and has recently been upgraded with motorized positioning, energy-dispersive monitoring and mechanical compatibility with laboratory XAS measurements [10].

Despite this progress, laboratory fluorescence-XAS at energies above approximately 20 keV remains challenging. Silicon-based detectors progressively lose effectiveness in the hard-X-ray range, whereas high-purity germanium detectors offer excellent spectroscopic performance but require cryogenic cooling. Conversely, high-Z compound semiconductors such as CdTe and CdZnTe provide high

hard-X-ray absorption efficiency and room-temperature operation, although their use requires careful control of charge transport, detector segmentation, electronic noise and spectral reconstruction.
As a consequence, the K-edges of technologically relevant elements such as Mo, Rh, Ag and Cd remain difficult to investigate by laboratory fluorescence-XAS under realistic experimental conditions. There is therefore a clear need for a compact platform combining high-resolution crystal optics with efficient, room-temperature hard-X-ray detection. Addressing this technological gap would extend laboratory XAS beyond 20 keV and establish a complementary capability between conventional laboratory instruments, synchrotron facilities and future compact photon sources.

2 Goals and Expected Outcome (2000 caratteri)

The **main goal** of the project is **to develop and validate a room-temperature laboratory platform for fluorescence-mode X-ray absorption spectroscopy (f-XAS) above 20 keV**. Source, crystal optics, sample geometry, CZT detection, electronics and simulations will be optimized as a single measurement chain to obtain chemically meaningful XANES spectra from samples for which transmission is impractical.

The project is organized into the following specific **goals (G)** and **expected outcomes (EO)**:

G1 – Establish a validated design and predictive model of the complete instrument.
EO1 – An integrated simulation workflow and a frozen experimental design, validated at the Ag K-edge and at one energy of at least 40 keV. Predictions and measurements will agree within 20% for detected rate and within 10% for spot size and energy resolution.

G2 – Develop and characterize a multichannel CZT detector operating at room temperature.
EO2 – A calibrated system with intrinsic efficiency above 90% at 40 keV, energy resolution of 1 keV FWHM or better over 20–40 keV, dead time below 10% and residual pile-up below 5% at the selected operating rate.

G3 – Validate the complete platform on reference compounds with different chemical states.
EO3 – Quantitative Ag K-edge XANES with effective XAS resolution of 10 eV FWHM or better, edge-position reproducibility within 1 eV, edge-step SNR of at least 20 and acquisition time below the 12 h laboratory benchmark. The normalized spectral deviation from reference data will be below 5%.

G4 – Demonstrate f-XAS on at least one thick, diluted or operando-compatible sample.
EO4 – Discrimination of at least two chemical states through an edge shift of at least 1 eV and/or a normalized white-line variation of at least 5%, with a significance of at least 3σ and an acquisition time below 12 h per spectrum. The measured operating domain will be reported in terms of concentration, sample geometry and acquisition time.

3 Methodology (10000 caratteri)

XAS measures the energy dependence of the linear absorption coefficient, μ(E), across an element-specific core-level absorption edge (see Figure 1). When the incident photon energy reaches the binding energy of a core electron, the absorption probability rises sharply and the resulting photoelectron probes both the unoccupied electronic states and the surrounding atomic structure. The near-edge region (XANES) is sensitive to oxidation state, orbital occupancy and local symmetry, whereas the oscillations above the edge (EXAFS) provide information on coordination numbers, interatomic distances and structural disorder. Laboratory XAS becomes increasingly demanding at hard-X-ray energies because the usable flux from conventional sources decreases and EXAFS requires accurate measurements over a broad energy interval. PRISM will use the Ag K-edge at 25.5 keV as its benchmark, first establishing a robust XANES protocol and then quantifying the EXAFS range achievable within laboratory-compatible acquisition times.

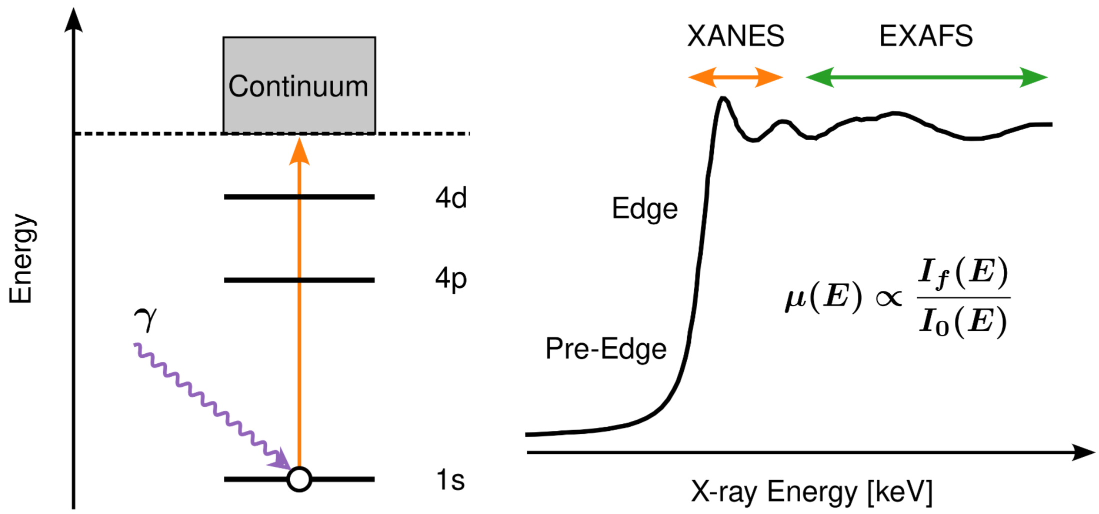

FIGURE 1. Principles of XAS illustrated for the Ag K-edge. Left: schematic of the electronic transitions involved in X-ray absorption. When the incident photon energy exceeds the Ag K-edge (≈25.5 keV), a 1s core electron is excited into the continuum, creating a core hole. Right: qualitative evolution of the absorption coefficient, μ(E), as a function of photon energy, highlighting the XANES (X-ray Absorption Near Edge Structure) and EXAFS (Extended X-ray Absorption Fine Structure) regions. XANES is primarily sensitive to the oxidation state and local electronic structure, while EXAFS provides quantitative information on the local atomic environment, including coordination numbers and interatomic distances.

The absorption coefficient μ(E) can be acquired in transmission or fluorescence mode. In a dispersive transmission arrangement, such as that being explored with VOXES, the broad source spectrum passes through the sample and a monochromator crystal spatially separates the transmitted energies onto a position-sensitive detector. Comparison of the spectra acquired with (I) and without (I_0) the sample provides μ(E). This approach is efficient but requires a homogeneous sample with an appropriate transmission thickness. PRISM will instead operate primarily in fluorescence mode. A crystal placed before the sample will select and focus one incident energy at a time, while an energy-resolving detector positioned approximately perpendicular to the incident beam will measure the characteristic fluorescence emitted following core-hole creation. For the Ag K-edge (see Figure 1), $I_f(E)$ will be extracted from the Ag Kα line complex at approximately 22.2 keV. This geometry extends XAS to thick, dilute, heterogeneous or geometrically constrained samples that are impractical in transmission. In the thin or dilute limit, after background, live-time and detector-response corrections, the absorption spectrum is proportional to the ratio $\mu(E) \propto I_f(E) / I_0(E)$ where $I_0(E)$ is the incident monochromatic intensity.

The incident energy, impinging on the sample, will be selected from the X-ray-tube continuum by Bragg diffraction on a cylindrically bent germanium and silicon crystal in a VOXES-derived von Hamos geometry. The rotation will be coordinated with the axis motors so that the source–crystal–sample geometry and the position of the focused spot are maintained throughout the scan. At each energy set point, a complete fluorescence spectrum will be acquired for a defined dwell time, and $I_f(E)$ will be obtained by fitting or integrating the selected fluorescence line after local-background subtraction. Finer energy steps will be used around the edge and coarser steps in the extended region, while the dwell time will be adapted to the predicted count rate and target statistical uncertainty.

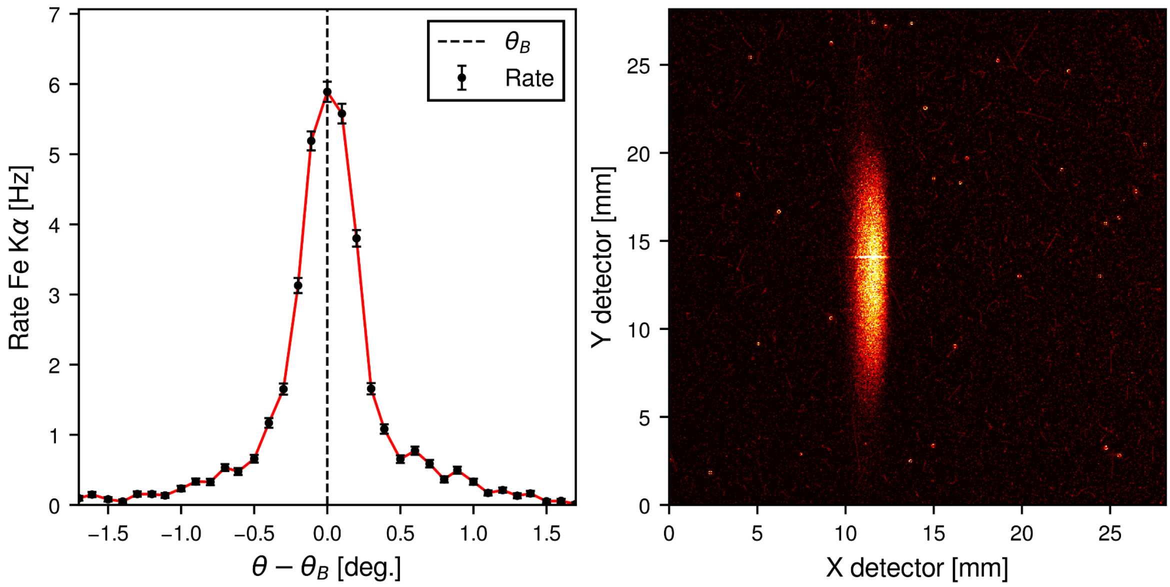

FIGURE 2. Performance of the VOXES setup in angular scanning and X-ray focusing. Left: angular dependence of the Fe Kα count rate measured using a mosaic graphite (002) crystal, showing a reflection profile extending over approximately 1°. Right: millimetre-scale Fe Kα focal spot recorded with a Timepix detector featuring a 28 x 28 mm^2 active area.

The feasibility of the required angular scan and focusing is supported by the existing VOXES experience. For instance, Figure 2 (left) shows the measured Fe Kα rate on the position-sensitive detector as the crystal is rotated through its nominal Bragg angle, demonstrating a controlled Bragg-angle scan. In Figure 2 (right), the same Fe Kα line is shown on a two-dimensional detector, exhibiting millimetric size. Although these measurements were performed in an X-ray-emission configuration and do not themselves constitute XAS, they validate two essential instrumental functions for PRISM: precise crystal rotation for energy selection and spatial focusing with a bent crystal. The project will build on this expertise, the available VOXES components and in-kind contributions, and the shielded laboratory infrastructure at LNF.

To overcome the declining detection efficiency of silicon detectors above 20 keV and the cryogenic requirements of germanium, PRISM will develop a custom multichannel CZT detector system. The design will build on the experience gained within SIDDHARTA, where CZT detectors were first tested in a collider environment, and will be specifically optimized for laboratory fluorescence-XAS.

Unlike commercial CZT detectors, the proposed architecture will be tailored to maximize the collection efficiency of the fluorescence emitted by the sample. A multichannel configuration will provide a larger active area and geometrical acceptance, while segmentation will distribute the counting rate across multiple channels, reducing channel-level pile-up and enabling localized background monitoring and rejection. The baseline design will employ 2-mm-thick CZT crystals with a channel pitch of 500 µm. The detector will be characterized in terms of absolute and relative efficiency, energy resolution, gain linearity and stability, dead time, and maximum usable count rate. The detector will be coupled to dedicated readout electronics capable of recording digitized waveforms for each channel. Waveform-based reconstruction will provide baseline correction, pulse-amplitude estimation, timing extraction, and channel equalization. Pulse-shape-analysis methods employing machine-learning methods, adapted from the experience gained with germanium detectors and currently being applied to CZT tests at BTF (Figure 5), will be used to identify and reject pile-up, saturation, electronic disturbances, and distorted pulses. The same methods will also help mitigate detector-specific effects such as incomplete charge collection, charge trapping, and charge sharing between adjacent channels. Energy calibration, gain matching, efficiency correction, and live-time correction will be performed independently for each channel before the individual spectra are combined into the final fluorescence spectrum.

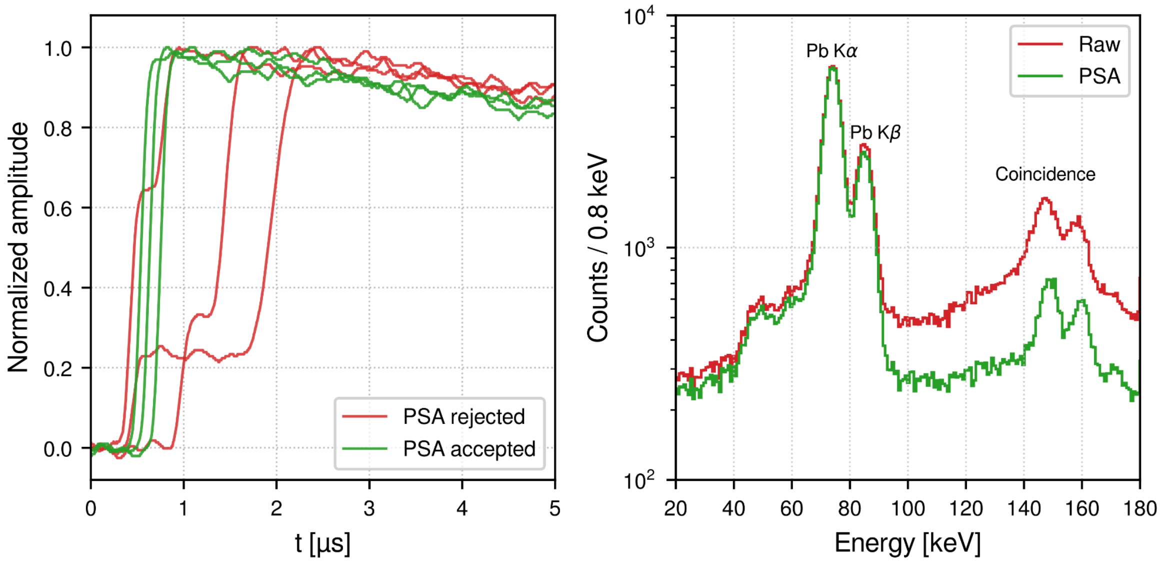

FIGURE 5. Machine-learning-assisted pulse-shape analysis (PSA) of CZT detector signals. Left: representative normalized waveforms accepted by the PSA as clean single-step events (green) and rejected multi-step events (red). Right: calibrated energy spectrum before (Raw) and after PSA selection, showing the preservation of the Pb Kα and Kβ fluorescence peaks and the suppression of the continuum and coincidence structures around 150–160 keV.

The incident intensity $I_0(E)$ will be measured with a motorized exchange system for the sample and CZT head, based on a long-travel linear axis and independent angular positioning. In the fluorescence configuration, the sample will be located at the monochromatic focus and the CZT array will observe it at approximately 90°. In the I_0 configuration, the sample will be translated out of the beam and the CZT detector will be moved and reoriented into the direct monochromatic beam at the focal position; the incident peak recorded by the CZT will then provide $I_0(E)$. Because the direct-beam rate will be much higher than the fluorescence rate, calibrated attenuators, shorter dwell times and, where necessary, reduced tube current or slit aperture will be used to prevent saturation and pile-up.

To determine the parameters of the components of the setup, within the allocated budget cost, an end-to-end, source-to-detector simulation of the spectrometer will be developed. The Ag K-edge at approximately 25.5 keV will be used as the reference case for defining the XANES measurement protocol and the required performance in terms of energy resolution, photon flux, signal-to-background ratio and acquisition time. The simulation workflow will consist of two coupled stages. First, SHADOW4 ray-tracing simulations building on the approach previously validated for the VOXES spectrometer, which will model photon transport from the X-ray tube through the collimating slits and crystal optics to the sample. Different silicon- and germanium-based crystal reflections will be investigated in the VOXES-type von Hamos geometry. Parametric studies will optimize the crystal material, reflection, curvature radius ρ_c , dimensions, source–crystal and crystal–sample distances, slit apertures and Bragg geometry. The simulations will quantify the geometrical acceptance, energy bandwidth, photon flux, beam profile and illuminated spot size at the sample, while also evaluating alignment and mechanical tolerances. The SHADOW4 output will then be transferred to GEANT4, which will simulate photon absorption in the sample, fluorescence emission, self-absorption, elastic and inelastic scattering, background contributions and energy deposition in the multichannel CZT detector. The combined simulations will be used in the first months of the project to optimize the source, crystal optics, slit system, sample geometry and detector configuration, balancing count rate, spectral resolution and background rejection. A coordinated motorized scanning scheme will also be defined to vary the incident energy across the Ag K-edge while maintaining the required source–crystal–sample geometry. The final outcome of this activity will be a design freeze specifying the X-ray source, crystal type and dimensions, curvature radius, distances, slit apertures, sample position, CZT geometry, motion stages and required positioning accuracy.

A proposed layout of the setup for PRISM is shown in Figure 5, highlighting all the components from the source, the slit system, crystal, sample and the custom CZT detector.

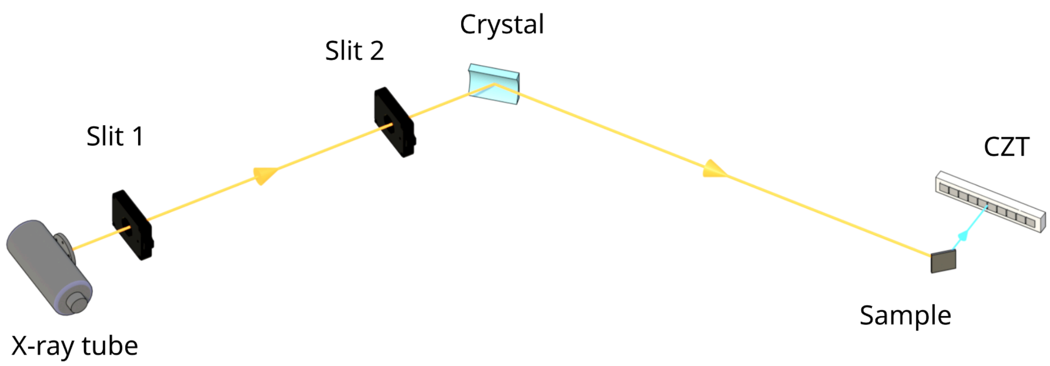

FIGURE 5. Conceptual layout of the proposed PRISM laboratory fluorescence-XAS setup. Radiation from the X-ray tube is collimated by two adjustable slits and monochromatized by a sagittally curved Ge(880) crystal. The diffracted beam illuminates the Ag sample, while the emitted fluorescence is collected off-axis by a linear multichannel CZT detector. The drawing is not to scale, and slit and detector positions are illustrative.

4 Organisation (4000 caratteri)

The PRISM project will have the following organization:

Name	Unit	FTE	Responsibility/WP
Simone Manti	INFN, Section of Frascati	1.0	Scientific Coordinator; overall coordination; WP1–WP4
Alessandro Scordo	INFN, Section of Frascati	0.3	Crystal optics, ray tracing, integration; WP1/WP3
Massimiliano Bazzi	INFN, Section of Frascati	0.3	Electronics/DAQ; WP2
Catalina Curceanu	INFN, Section of Frascati	0.1	Scientific review and application validation; WP4
Totale		**1.7**	**-**

It will be divided in the following Work Packages (WP):

WP1 - Requirements, simulations and design freeze (M1-M6)
d
WP2 - CZT, front-end, DAQ, calibration and characterization (M4-M15)
d
WP3 - Procurement, mechanics, integration and commissioning (M10-M18)
d
WP4 - Ag reference samples and realistic sample demonstration (M16-M24)

The project **Milestones** are:

- **MS1 – Instrument design frozen (M6).** Experimental requirements, complete source–optics–sample–detector geometry and procurement specifications approved.
- **MS2 – CZT detection chain validated (M15).** Intrinsic efficiency above 90% at 40 keV, energy resolution of 1 keV FWHM or better, dead time below 10% and residual pile-up below 5% at the selected operating rate.
- **MS3 – Integrated instrument commissioned (M18).** Synchronized scanning and energy calibration operational; measured rate within 20%, and spot size and energy resolution within 10% of simulation predictions.
- **MS4 – Fluorescence XAS capability demonstrated (M24).** Ag K-edge XANES obtained with SNR $\geq$ 20 and edge-position reproducibility within 1 eV; two chemical states discriminated in a realistic sample with a significance of at least 3σ; operating domain documented.

The project **Deliverables** (D) are:

- **D1** – Validated SHADOW4–GEANT4 workflow, frozen instrument design and procurement specifications (M6).
- **D2** – CZT calibration and performance-characterization dataset with acceptance report (M15).
- **D3** – Ag foil and reference-compound XANES dataset with quantitative validation report (M21).
- **D4** – Realistic-sample dataset, operating-domain assessment, documented analysis tools and final project report (M24).

The Work Packages and the Milestones are summed in the Gantt table in Figure 6:

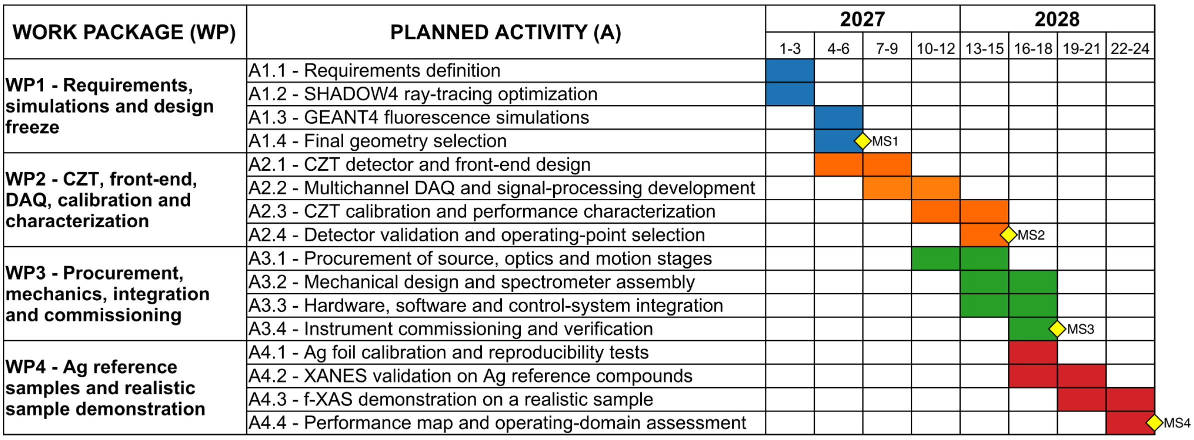

FIGURE 6. Gantt table of the PRISM objected divided in each Work Packages (WPs) and for each one its Planned Activity (A) through the two years and Milestones (MS) are indicated by the yellow mark.

5 Synergies with Other Activities (3000 caratteri)

PRISM is embedded in complementary INFN activities while retaining independent objectives, budget and deliverables.

At LNF, PRISM builds directly on VOXES, developed through an INFN-CSN5 Young Researchers Grant. It will reuse established know-how and, where available, instrumentation for bent-crystal optics, alignment, mechanics, X-ray-source operation and spectral analysis. The LNF Directorate has committed access to laboratory resources and instrumentation. This institutional contribution is in kind; no separate cash co-funding is included.

Cross-committee synergy arises from SIDDHARTA-2/KAONNIS (CSN3), through shared expertise in precision X-ray detection, semiconductor-detector characterization, low-noise readout, calibration, dead-time and pile-up control, and spectral reconstruction. This expertise will support the PRISM CZT system, while PRISM will return methods relevant to hard-X-ray detector characterization. No CSN3 financial contribution is assumed.

PRISM is also strategically complementary to EuAPS, the INFN-led PNRR project involving CNR and the University of Rome Tor Vergata and developing a betatron X-ray user facility at SPARC_LAB. After laboratory validation, PRISM will use its measured source and detector requirements to assess transferability to this compact pulsed source and, if operating conditions

permit, define a pilot test. EuAPS funding and resources are not included in the PRISM budget, and EuAPS availability is not required to achieve the project deliverables.
Rigaku and Quantum Design have been approached regarding the scientific and technological relevance of laboratory fluorescence-XAS above 20 keV. If the letters of support are signed, their role will be limited to following the project and exploring future technical exchanges and applications. No industrial cash or in-kind co-funding is currently committed, and any procurement will follow INFN procedures without preferential treatment.

6 Impact and Spin-offs (3000 caratteri)

PRISM will establish within INFN a new laboratory capability for quantitative fluorescence-XAS above 20 keV using room-temperature detection, a regime where efficient non-cryogenic spectroscopy remains challenging. Its impact will extend beyond one instrument: integrating crystal optics, multichannel CZT, low-noise electronics and DAQ with SHADOW4-GEANT4 modelling will consolidate at LNF expertise in semiconductor detectors, precision X-ray spectroscopy, radiation transport and spectral reconstruction. The platform, validated models, calibration protocols and performance maps will be reusable by INFN activities developing hard-X-ray instrumentation. PRISM will valorize existing VOXES infrastructure and strengthen INFN's capacity to translate detector R&D from simulation and subsystem tests into validated instruments. After laboratory validation, its design rules may also inform EuAPS. The project will provide interdisciplinary training for young researchers and technical staff.
For the wider scientific community, PRISM addresses the limited availability of laboratory fluorescence-XAS at the K-edges of technologically relevant elements such as Mo, Rh, Ag and Cd. Meeting its quantitative targets will enable routine screening, method development, sample and cell optimization and long in-situ or operando studies on thick, dilute and heterogeneous specimens that are difficult to measure in transmission. This capability will complement, not replace, synchrotrons and XFELs: preliminary measurements and protocol optimization can make scarce beamtime more effective, while repeated scans and slowly evolving processes can remain in the laboratory. Beneficiaries span catalysis, electrochemistry, energy materials and advanced alloys. Documented simulation workflows, calibration methods, reference datasets and operating-domain maps will promote reproducibility and guide other laboratories in assessing and adapting the approach.
Potential technology-transfer outcomes include the modular room-temperature CZT detection chain—front-end electronics, multichannel DAQ, calibration and pulse-reconstruction methods—the integrated digital model of source, optics, sample and detector, and the compact hard-X-ray spectrometer architecture. Following experimental validation, these assets could be assessed with INFN technology-transfer structures for protectable know-how, licensing or industrial co-development. Applications may include routine materials characterization, quality control and long-duration monitoring of catalysts, batteries and advanced materials. Technical exchange with X-ray-instrumentation companies, component suppliers and simulation-oriented INFN spin-offs could accelerate uptake. Industrial adoption will remain conditional on demonstrated performance, robustness, manufacturability, cost and market need.

7 Costs

The financial request is structured according to the technical development of the project and distributed over the two years of activity. During Year I, the budget is mainly devoted to the development of the fluorescence detection system, including the multichannel CZT detector, the digitizer and the bent Ge/Si crystal optics required for the selected absorption edges. These components are essential for the design, integration and initial commissioning activities carried out within WP2. The Year II request supports completion of the experimental platform through the acquisition of the X-ray tube and its power supply, motorized translation and rotation stages, slits, custom mechanical supports for the sample and detector, and reference targets for the validation campaign. Travel funds are reserved for experimental activities, collaboration meetings and dissemination of the results at an international conference, subject to acceptance of an oral contribution. All equipment items and all individual expenses exceeding €2,000 will be supported by detailed supplier quotations. Costs below this threshold are based on preliminary supplier or internal estimates. The requested amounts remain below the maximum annual funding limit and are directly linked to the activities and deliverables described in the corresponding Work Packages.

Cost category	Item / description	Year I (€)	Year II (€)	Total (€)
Equipment	Multichannel CZT detector	30000	0	30000
Equipment	Digitizer	20000	0	20000
Equipment	Bent crystal optics	15000	15000	30000
Consumables	Reference samples and targets	0	2000	2000
Equipment	Precision slits	0	2000	2000
Services	Custom sample and CZT supports	0	3000	3000
Equipment	Motorized translation stages	0	5000	5000
Equipment	Motorized rotation stages	0	3000	3000
Equipment	X-ray tube	0	7000	7000
Equipment	X-ray tube power supply	0	10000	10000
Equipment	Closed-loop cooling system	0	5000	5000
Consumables / services	Consumables and precision machining	0	4000	4000
Travel	Experimental activity at collaborating facilities	0	2000	2000
Travel	Project and collaboration meetings	0	1500	1500
Travel	EXRS 2028 conference (oral presentation only)	0	3000	3000
Total requested		65000	62500	127500

8 Risk Assessment

X-ray source, crystal optics and motorized stages

- Risk: Delivery or installation delays, or components failing interface or performance specifications.
- Consequence: Postponed assembly and commissioning, with less time for WP4 and potentially reduced flux, resolution or scan range.
- Mitigation: Define interfaces and acceptance criteria during WP1; order long-lead components immediately after the design freeze; identify compatible alternatives; perform incoming acceptance tests; use available VOXES components for preliminary tests where compatible.

CZT detector, front-end electronics and multichannel DAQ

- Risk: Component delays, failed channels, excessive noise, charge trapping, non-uniform response, dead time or pile-up.
- Consequence: Delayed detector validation and degraded useful count rate or fluorescence/background discrimination.

- Mitigation: Test the system progressively at channel and module level; optimize bias, grounding, shielding, shaping and operating rate; apply channel equalization, pulse-shape analysis and dead-time corrections. A validated sub-array may support early integration without relaxing the final acceptance criteria.

VOXES/LNF infrastructure and technical services

- Risk: Temporary unavailability or malfunction of the experimental area, X-ray source, shielding, interlocks, workshop or technical services.
- Consequence: Interruption of assembly, calibration or data taking.
- Mitigation: Schedule access and safety reviews in advance; perform preventive checks; include schedule contingency and staged bench tests. During downtime, continue simulations, electronics tests, control-software development, mechanical preparation and data analysis.

Crystal-optics and scanning system

- Risk: Crystal performance differs from predictions, or positioning repeatability, backlash or synchronization is inadequate.
- Consequence: Incident-energy resolution, calibration or scan reproducibility falls below the XANES requirements.
- Mitigation: Perform tolerance studies before procurement; define supplier metrology and bench-acceptance tests; retain adjustable mounts, slits and distances and an alternative crystal/reflection option; recalibrate the energy-position relation using reference absorption features.

End-to-end model and integrated performance

- Risk: Measured flux, background, self-absorption or CZT response differs from SHADOW4–GEANT4 predictions, resulting in insufficient SNR or excessive acquisition time.
- Consequence: A suboptimal design may be frozen, or G3/G4 may not be achieved in the initial configuration.
- Mitigation: Propagate uncertainties and validate each simulation stage against subsystem measurements; identify experimental losses and background sources; optimize source operation, slits, geometry, detector solid angle, shielding and reconstruction. Complete Ag-reference validation before proceeding to WP4.

Ag references, realistic sample and sample environment

- Risk: Reference materials are delayed, or the selected sample is unstable, insufficiently reproducible or affected by excessive attenuation, self-absorption or contamination.
- Consequence: Delayed WP4 measurements or ambiguous chemical-state discrimination.
- Mitigation: Procure and characterize Ag standards early; preselect alternative concentrations, thicknesses and geometries. If an operando-compatible environment is unavailable, use a controlled thick, diluted or heterogeneous static sample preserving the same chemical observable.

9 References

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